

Helix 3000

Water Quality Logger

Operator Manual

Document HX3-OM-07. Applies to firmware 4.2 and later.

Manufactured by Meridian Systems Ltd, Bellhaven.

1. Specifications

Parameter	Value
Measured channels	pH, turbidity, conductivity, temperature
pH range	2.0 to 12.0, accuracy +/- 0.05
Turbidity range	0 to 1,000 NTU, accuracy +/- 2%
Conductivity range	5 to 50,000 uS/cm
Temperature range	-5 to 60 C
Logging interval	30 seconds to 24 hours, configurable
Internal storage	512,000 records
Battery life	18 months at 15-minute logging
Ingress protection	IP68 to 10 metres
Operating temperature	-10 to 55 C
Weight	1.4 kg including battery

2. Safety

The logger contains a lithium thionyl chloride battery. Do not incinerate, crush or short the battery pack. Replace only with the Meridian part HX3-BAT-2; third-party cells of the same nominal voltage have different discharge characteristics and will cause premature low-voltage shutdown.

The instrument is not certified for use in explosive atmospheres.

3. Installation

3.1 Mounting

Mount the logger vertically with the sensor guard downward. Horizontal mounting traps air against the turbidity window and produces intermittent high readings that are easily mistaken for a genuine turbidity event.

Maintain at least 150 mm of clearance below the sensor guard. Where the logger is deployed in a channel, position it at least one metre downstream of any bend or obstruction so that the flow at the sensor is not turbulent.

3.2 Cable and strain relief

Support the cable with the supplied clamp within 300 mm of the connector. The connector is rated to IP68 but is not rated for load; suspending the instrument by its cable will eventually breach the seal.

Torque the connector collar to 2.5 Nm. Do not use tools; the collar is designed for hand tightening and over-torquing will deform the O-ring seat.

3.3 First power-up

Insert the battery pack and wait for the status LED to show a steady green. A flashing amber LED at power-up indicates the real-time clock has lost its setting and the instrument must be connected to the configuration software before deployment, or records will carry an incorrect timestamp.

Confirm the logging interval before deployment. The factory default is 15 minutes; a unit returned from service is set to 60 minutes.

4. Calibration

4.1 Schedule

Channel	Interval	Standard required
pH	30 days	Two-point, buffer 4.0 and 7.0
Turbidity	90 days	Formazin 20 NTU and 800 NTU
Conductivity	180 days	Single point, 1,413 uS/cm
Temperature	Factory only	Not field calibratable

4.2 Procedure

Allow the standard and the sensor to reach the same temperature before calibrating. A temperature difference of more than 3 C between standard and sensor is the most common cause of a calibration that appears to succeed but drifts within days.

Rinse the sensor with deionised water between standards. Do not wipe the pH bulb; abrasion damages the glass membrane and shortens sensor life considerably.

4.3 Calibration records

The instrument stores the last 40 calibration events with timestamp, standards used and the resulting slope. A pH slope below 85 percent indicates the sensor is near end of life and should be replaced rather than recalibrated.

5. Troubleshooting

Code	Meaning	Action
E01	Battery below 3.2 V	Replace battery pack
E02	Sensor not detected	Check connector, reseal, inspect pins
E03	pH slope out of range	Recalibrate; replace sensor if below 85%
E04	Turbidity window fouled	Clean window per section 6.2
E05	Storage full	Download and clear records
E06	Clock not set	Connect configuration software
E07	Internal humidity high	Seal breached; return for service
E08	Firmware image invalid	Re-flash from known good image

5.1 Readings that drift upward over days

Gradual upward drift on the turbidity channel almost always indicates biofouling of the optical window rather than a genuine change in water quality. Compare against a manual grab sample before treating the trend as real.

5.2 Intermittent spikes

Isolated single-record spikes are usually air bubbles passing the sensor. If spikes appear at a consistent time of day, check for a pump cycling upstream. Persistent spikes with no upstream cause suggest the instrument is mounted horizontally or with insufficient clearance; see section 3.1.

5.3 Unit will not communicate

Confirm the configuration cable is the HX3-CBL-3, not the older HX3-CBL-1 supplied with the Helix 2000. The older cable fits mechanically but does not carry the data pair on the same pins.

6. Maintenance

6.1 Routine inspection

Inspect the sensor guard, cable and connector every 90 days in normal deployments and every 30 days where the water carries heavy sediment or the instrument is deployed in a wastewater stream.

6.2 Cleaning

Clean the optical window with the supplied soft brush and a dilute detergent. Do not use solvents, acetone or abrasive pads. For heavy biofouling, soak the sensor guard in a 5 percent acetic acid solution for 20 minutes, then rinse thoroughly with deionised water.

6.3 Storage

Store the pH sensor wet, in the supplied storage solution, never in deionised water. Storing a pH sensor dry or in deionised water will destroy the reference junction, typically within a week, and the damage is not recoverable by recalibration.

For storage longer than six months, remove the battery pack to avoid the risk of leakage.

7. Warranty

The instrument body is warranted for 24 months from the date of delivery. Sensors are consumable items and are warranted for 6 months. The battery pack is not covered.

The warranty does not cover damage from deployment beyond the rated depth, use of a non-Meridian battery pack, or breach of the connector seal caused by suspending the instrument by its cable. Warranty service requires the original delivery reference; units returned without one are handled as chargeable repairs.